An Introduction to Dynamics

SPH-3UI Grade 11 Physics

April 24, 2026

Why study dynamics?

Kinematics studies **how** objects move. It answers questions like how far, how fast, and for how long motion occurs.

Dynamics studies **why** objects move the way they do. It focuses on the causes of motion — what forces act, why something speeds up, slows down, or changes direction.

What is Force?

A force is a **push** or a **pull** that can cause an object to change its motion or shape.

In physics, force is represented by a **vector** because it has both magnitude and direction. For example, pushing a book sideways will have a different effect than pulling it upward.

Forces are the fundamental way that matter interacts with matter.

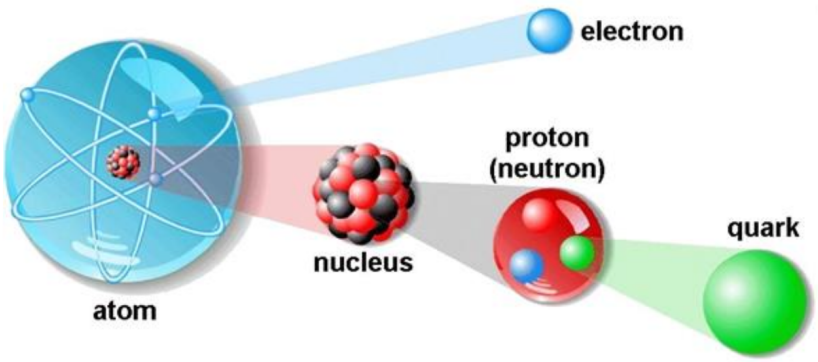
Measuring Forces: The Newton

The SI unit of force is the **newton (N)**, named after Isaac Newton. One newton is the force required to accelerate a 1 kg mass by 1 m/s^2.

Tools to measure force:

- A spring scale stretches more with greater pulling force.
- A digital force sensor records and graphs force changes over time.

Atoms and Fundamental Particles



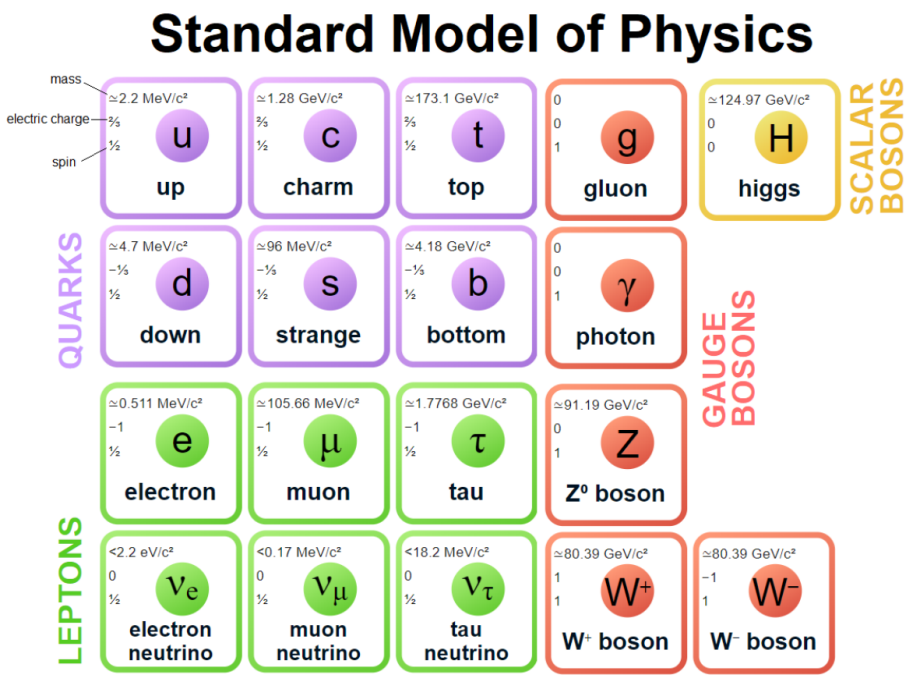
Atoms are made of protons, neutrons, and electrons. The electron is a fundamental particle called a lepton. Protons and neutrons are made of smaller particles called quarks:

- Proton: 2 up quarks + 1 down quark.
- Neutron: 2 down quarks + 1 up quark.

The Standard Model

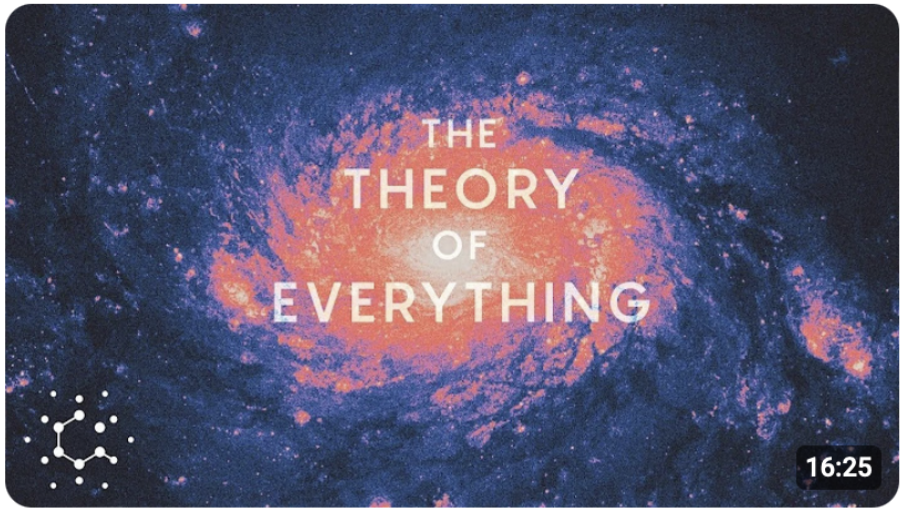
The **Standard Model of Particle Physics** is our best theory for explaining the fundamental building blocks of the universe.
It divides all particles into two main types:

- **Fermions:** particles that make up matter (quarks and leptons).
- **Bosons:** particles that carry forces (photons, gluons, W and Z bosons, etc.).



Watch These!

The Standard Model: A Triumph of Science



The Standard Model of Particle Physics: A Triumph of Science

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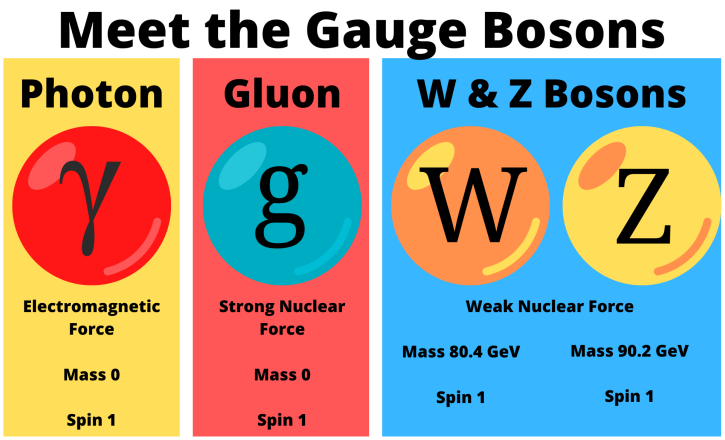
How Do Particles Exert Forces?

Forces occur through the exchange of **gauge bosons**, the messenger particles of nature.

When two electrons repel each other, they're constantly exchanging photons. This transfer of momentum is what we experience as a force.

Imagine two people in rolling chairs tossing a medicine ball. Each slides backward when the ball is thrown or caught. The ball acts like a photon, carrying the force between them.



The Four Fundamental Forces

All interactions in the universe can be explained by four fundamental forces:

- **Gravitational Force:** acts between masses; always attractive.
- **Electromagnetic Force:** acts between charged particles; can attract or repel.
- **Strong Nuclear Force:** binds protons and neutrons inside the nucleus.
- **Weak Nuclear Force:** responsible for radioactive decay and particle transformations.

Type of force	Approximate relative strength	Range	Effect
gravitational	1	infinite	attract only
electromagnetic	10^{20}	infinite	attract and repel
strong nuclear	10^{38}	less than 10^{-15} m	attract and repel
weak nuclear	10^{25}	less than 10^{-18} m	attract and repel

Why the Standard Model is Incomplete

Although extremely successful, the Standard Model does not explain everything. It leaves out gravity, dark matter, and dark energy.

Some unanswered questions:

- Why are there three families of quarks and leptons?
- Why do particle masses vary so much?
- What gives particles their mass? (Answered in 2012!)
- How does gravity fit with the other forces?

Scientists are searching for a **Grand Unified Theory (GUT)** or even a **Theory of Everything (TOE)**.

Everyday Forces

Example: A wagon is being pushed and pulled by two children.

Which forces are acting on the wagon?

Everyday Forces

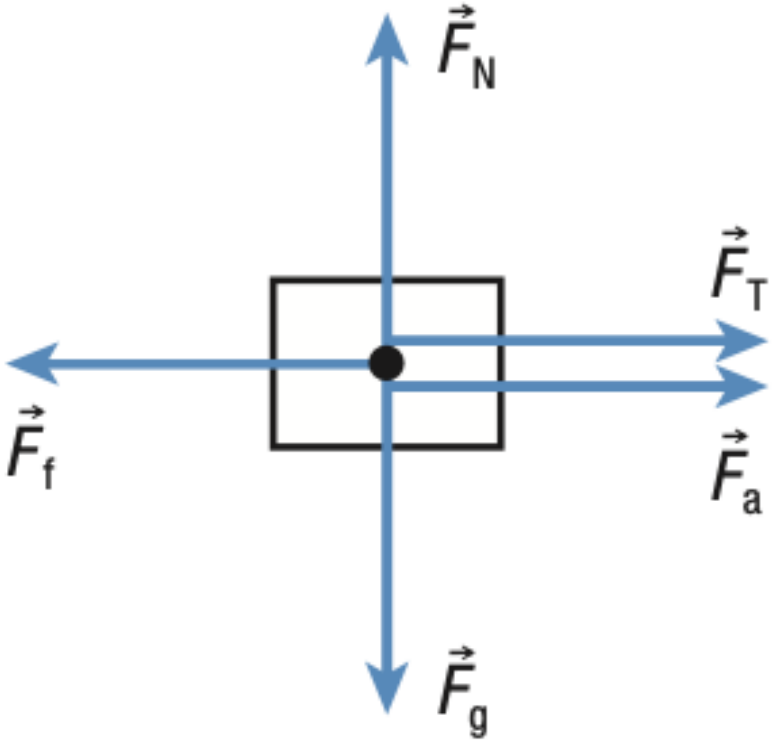
Objects experience many familiar forces:

- **Applied Force ($\vec{F}_A$):** push or pull from contact.
- **Tension ($\vec{F}_T$):** pull from a rope or string.
- **Normal Force ($\vec{F}_N$):** upward support from a surface.
- **Friction ($\vec{F}_f$):** opposes motion along a surface.
- **Gravity ($\vec{F}_g$):** attraction due to mass.

Free-Body Diagrams (FBDs)

To analyze motion, we draw a **free-body diagram** (FBD). It shows an object and all the forces acting on it.

- Draw the object as a box or dot.
- Represent each force with an arrow pointing away from the object.
- Longer arrows represent larger forces.
- Label each force ($\vec{F}_N$, $\vec{F}_g$, etc.).



Net Force

Once all forces are identified, they combine to form a single **net force**.

$$\vec{F}_{net} = \sum \vec{F}$$

The net force determines the object's acceleration:

$$\vec{F}_{net} = m\vec{a}$$

If the net force is zero, the object remains at rest or moves at constant velocity.

Example 1: Free-Body Diagrams (FBDs)

- Draw the object as a box or dot.
- Represent each force with an arrow pointing away from the object.
- Longer arrows represent larger forces.
- Label each force ($\vec{F}_N$, $\vec{F}_g$, etc.).

Try the following examples:

- A cup sitting at rest on a table.
- A large trunk in the basement is pulled by a rope tied to the right side of the trunk by a person. The trunk does not move.
- A baseball player is sliding to the left across the ground.
- A desk is pushed to the left across the floor.